

DIGITAL FORENSICS INVESTIGATION REPORT

Volatility 3 Memory Forensics Analysis

1. Case Information

Case ID	CASE-2026-004
Lead Investigator	Cyber Analyst Team 404
Target Memory Image	mem_dump_cybersecurity_incident.raw
Operating System Profile	Win10x64_19041 (Windows 10 Pro)
Report Generated	2026-07-04 15:37:04

2. Executive Summary

This report documents the results of a memory forensics examination performed on the target memory image **mem_dump_cybersecurity_incident.raw**, acquired from a system running **Win10x64_19041 (Windows 10 Pro)**. Automated analysis of the process list, DLL list, network connections, and command-line history identified **3 findings** of forensic interest, resulting in an overall risk classification of **HIGH** (Risk Score: 80/100 (CRITICAL LEVEL)). Based on these findings, **4 recommendations** have been provided in Section 5 of this report to guide remediation and any further investigation.

3. Risk Assessment

Overall Risk Score	80/100 (CRITICAL LEVEL)
Risk Classification	HIGH

Detection Category Breakdown

Detection Category	Items Detected
Hidden / Unlinked Processes	1
Unknown or Unsigned DLLs	4089
Suspicious External Network Connections	43

4. Detailed Findings

The table below lists each forensic indicator identified during automated analysis of the memory image, in the order it was detected.

#	Finding
1	Hidden Process detected (1 found) [+40]
2	Unknown DLL detected (4089 found) [+20]
3	External Connection detected (43 found) [+20]

5. Recommendations

The following mitigation and follow-up actions are recommended based on the findings above.

#	Recommended Action
1	Isolate the affected host from the network and terminate the hidden process(es) immediately; process hiding indicates likely rootkit or DKOM-based malware.
2	Extract and sandbox the flagged DLL(s) for static/dynamic analysis; verify digital signatures and hashes against known-good baselines and threat intelligence feeds.
3	Block outbound traffic to the flagged external IP(s) at the firewall/proxy and review DNS and connection logs for the same destination across other hosts.
4	Preserve the memory image and all Volatility3 output files under chain-of-custody procedures for potential further analysis.

6. Scope and Methodology

Analysis was performed on the memory image **mem_dump_cybersecurity_incident.raw** using the Volatility 3 memory forensics framework, with the OS profile **Win10x64_19041 (Windows 10 Pro)**. The examination reviewed process listings (pslist/psscan), loaded modules (dlllist), active and historical network connections (netscan), and process command-line arguments (cmdline) to identify indicators of process hiding, unsigned or unrecognized DLLs, suspicious external network activity, and encoded or obfuscated PowerShell usage. Findings were scored and classified by the automated risk-scoring engine described in this report's supporting documentation.

7. Report Disclaimer

This report contains confidential information derived from digital forensic analysis and is intended solely for authorized personnel involved in this investigation. Unauthorized access, use, or distribution of this document is strictly prohibited. Findings are based solely on the artifacts available at the time of memory acquisition and should be corroborated with other evidence sources where possible.